Isa Vialard

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Education

2021 – 2024 **Ph.D. at LMF, ENS Paris-Saclay**, Gif-sur-Yvette
Thesis title: *Measuring Well Quasi-Orders and the complexity of verification,* under supervision of Philippe Schnoebelen

2020 – 2021 **Master Parisien de Recherche en Informatique**, Paris.

2017 – 2021 **École Polytechnique**, Palaiseau.

Experiences

2024-2025 **Postdoctoral researcher at the Max Plank Institute for Software Systems, in the Foundations of Algorithmic Verification group led by Joël Ouaknine.**
Topic : *Weighted Timed Games.*

2021 **Research internship with Philippe Schnoebelen**, ENS Paris-Saclay
Topic: *Measuring the width of Cartesian product of Well Quasi-Orders.*

2020 **Research internship with Patrick Totzke**, Liverpool University
Topic: *(Bi-)Simulation between Gap-order Constraint Systems.*

2019 **Internship at AdaCore with Yannick Moy**, Paris
Topic: *Ada/Spark binding for C crypto libraries TweetNaCl and Libsodium.*

Publications

Journal articles

Order — I. Vialard, *On the width of the cartesian product of ordinals,* Order, vol. 42, pp. 37–58, Apr. 2024. doi: 10.1007/s11083-024-09668-8.

Acta Informatica — Ph. Schnoebelen and I. Vialard, *On the piecewise complexity of words,* Acta Informatica, vol. 62, 2025. doi: 10.1007/s00236-025-00480-4.

Logic Journal of the IGPL — I. Vialard, *On maximal order type of the lexicographic product,* Accepted for publication in Logic journal of the IGPL. arXiv: 2409.09699 [math.LO].

Conference proceedings

FOSSACS 2026 — Q. Guilmant, J. Ouaknine, and I. Vialard, *Undecidability of the value problem for 2-clock weighted timed games,* **distinguished paper** at FOSSACS 2026.

CSL 2026 — I. Vialard, *Deciding the Value of Two-Clock Almost Non-Zeno Weighted Timed Games*, accepted at CSL 2026.

AAAI 2026 — R. Aghamov, C. Baier, J. Ouaknine, J. Piribauer, M. Vahanwala, I. Vialard, *Temporal Properties of Conditional Independence in Dynamic Bayesian Networks,* accepted at AAAI 2026.

Publications (continued)

ICALP 2025 — Ph. Schnoebelen, J. Veron, and I. Vialard, *A tropical approach to the compositional piecewise complexity of words and compressed words,* in ICALP 2025

SOFSEM 2024 — M. Praveen, P. Schnoebelen, J. Veron, and I. Vialard, *On the piecewise complexity of words and periodic words,* in SOFSEM 2024: Theory and Practice of Computer Science, LNCS, 2024, pp. 456–470. doi: 10.1007/978-3-031-52113-332.

MFCS 2023 — I. Vialard, *Ordinal measures of the set of finite multisets,* in Mathematical Foundations of Computer Science (MFCS 2023), LIPIcs. doi: 10.4230/LIPIcs.MFCS.2023.87.

In preparation

- M. Blondin, M. Cadilhac, B. Courchesne, L. Guillou, C. Mascle, I. Vialard, Population Protocols over Ordered Agents, submitted for publication.
- S. Abriola, S. Halfon, A. Lopez, S. Schmitz, P. Schnoebelen, and I. Vialard, *Measuring well quasi-ordered finitary powersets,* submitted for publication. url: https://arxiv.org/abs/2312.14587

Scientific talks

Conference talks

CSL 2026 — **EACSL Annual Conference on Computer Science Logic** , Paris 2026.
Title: *Deciding the Value of Two-Clock Almost Non-Zeno Weighted Timed Games.*

Highlights 2025 — **Highlights of Logic, Games and Automata,** Saarbrucken 2025.

ICALP 2025 — **International Colloquium on Automata, Languages, and Programming**, Aarhus 2025.
Title: *A tropical approach to the compositional piecewise complexity of words and compressed words*

MFCS 2023 — **International Symposium on Mathematical Foundations of Computer Science**, Bordeaux 2023.
Title: *Ordinal Measures of the Set of Finite Multisets*

Highlights 2022 — **Highlights of Logic, Games and Automata,** Paris 2022.

Workshops

Logic and Mentoring Workshop **Paris** (2026). Highlights collaborative research week, **Saarbrucken** (2025). Autoboz research week, **Aahrus** (2025). Workshop *Words and Subwords*, **Liège** (2025). Workshop *WQO-BQO - what is up?*, **Lyon** (2023). Research school *Discrete mathematics and logic,* **Marseille**(2023).

Invited seminars

Seminar 68NQRT of IRISA, **Rennes** (2025). Seminar MTV of LaBRI, **Bordeaux** (2025). Seminar Informel of LMF, **Paris-Saclay** (2025). MoVe Seminar of LIS, **Marseille** (2025). Automata Seminar of IRIF, **Paris** (2025). Journées annuelles du GT Verif, **Lille** (2024). Seminar LX of LaBRI, **Bordeaux** (2024). Automata Seminar of IRIF, **Paris** (2024). Journées annuelles du GT Verif, **Paris** (2023). Seminar at Liverpool University, **Liverpool** (2023).

Teaching (for a total of 198h)

2021 – 2024 — **Formal Languages**, ENS Paris-Saclay L3.

Teaching (for a total of 198h) (continued)

2023 – 2024 **Logic Project (Introduction to Coq and Agda)** , ENS Paris-Saclay L3.

2022 – 2023 **Architecture and systems** , ENS Paris-Saclay L3.

2021 – 2022 **Rewriting theory** , ENS Paris-Saclay M1.

Supervision

2026 Supervising Angel Wuttke during his internship on the subject of parameterized timed automata.

2025 Supervising Angel Wuttke during his internship on the subject of two-clock weighted timed games with negative weights.

Skills

Languages	Fluent in French and English.
Coding and proof assistants	CaML, Python, C, Coq, Agda, Why3.